

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*.

This session is not a board session. Both derivations are step-slide runs — one line per surface, built in front of the room — so the board carries exactly **two lines** and nothing else. That is the change from session 4, which spent thirty-two minutes at the board, and it is deliberate: a derivation that has to be legible in the PDF afterwards belongs on a slide, and a quantity that has to stay visible for seventy minutes belongs on the board.

Conventions. ASK — put it to the room before writing, and the answer to expect. **POINT** — the moments you turn and point at the ledger rather than the screen. **IF ASKED** — the question that always comes.

The board plan

Same room as sessions 3 and 4: blackboard across the front, screen down in the **centre**, a **left wing** and a **right wing**.

LEFT WING — the ledger. Two lines, written as they are derived, never wiped.

RIGHT WING — stays empty. If you find yourself working on it, you have left the slides, and the room is now watching an unrecorded derivation. Go back.

Line 1 — at minute 16, during derivation 1

Written at step 4 of the first step-slide run, while the slide showing it is up.

$$p^* = \frac{\alpha}{\gamma + \mu} \quad \mu = \frac{\ln 2}{T_d} = \frac{\ln 2}{30} = 0.023 \text{ min}^{-1}$$

ASK before writing μ : *what is the doubling time of the host we have been using all term?* — 30 minutes. It is retrieval question 3, and they should have it.

Say while writing: this is a **balance**, not an equilibrium. Nothing has stopped. Production and removal are equal and both are still running.

IF ASKED *why is dilution a rate at all — nothing is destroyed?* The cell divides and the pool is shared between two daughters, so the concentration in each is halved on the division timescale. Per molecule, per unit time, that is indistinguishable from degradation, which is why the two terms add.

Line 2 — at minute 31, during derivation 2

Written directly **under** line 1, at step 3 of the second run.

$$t_{1/2} = \frac{\ln 2}{\gamma + \mu}$$

POINT, and say it: α is not in this expression. That settles the vote you took at minute 20 — **B was right**, and the argument is now on the board rather than in anyone's opinion.

Then, without writing anything more, point at both lines together: same denominator. One knob sets the level alone; the other sets the level *and* the speed. That is the whole session, and it is two lines.

Where you point at the ledger later

MINUTE	WHAT IS ON SCREEN	WHY THE LEDGER HAS TO STILL BE THERE
40–44	Andersen 1998, Fig. 3A	his half-lives are γ alone; the ledger's μ is what you add to them
50–58	handout, items 1 and 2	item 2 is $k - \mu = \gamma$, read straight off line 1
62–66	the speed–level trade	the curve is $\mu/(\gamma + \mu)$ against $\ln 2/(\gamma + \mu)$ — both lines, plotted
66–76	pose · vote · argue	the design question is which of the two knobs to turn

Erase the ledger and the last thirty minutes stop making sense. If a previous class has left the boards full, clear the left wing before the room fills, not at minute 16.

The two numbers to have in your head

Not for the board — for answering out loud without reaching for a slide.

- $\mu = 0.0231 \text{ min}^{-1}$ at $T_d = 30 \text{ min}$. An **undegradable** protein still has a **30 minute** half-life. That is the floor, and it is the number the whole session is built to make surprising.
- The handout's two fits: LAA gives $k = 0.041$, $t_{1/2} = 17 \text{ min}$; AAV gives $k = 0.035$, $t_{1/2} = 20 \text{ min}$. Subtracting μ from the second returns **59 min**, which is Andersen's published 60. If a student asks whether the data are real: they are synthetic, drawn from Andersen's numbers, and the handout says so in italics.